ORIGINAL ARTICLE

Association of fasting glucose levels with incident atherosclerotic cardiovascular disease: An 8-year follow-up study in a Chinese population

Highlights
- Diabetes other than impaired fasting glucose was associated with increased risk of incident ASCVD after 7.8 years follow-up among 18 610 participants.
- Impaired fasting glucose increased the risk of developing diabetes.
- Prevention strategy on reducing incidence of diabetes in persons with impaired fasting glucose is critical to improve cardiovascular health status.

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Abstract

Background: Identification of the population at high risk of developing atherosclerotic cardiovascular disease (ASCVD) is critical for its prevention. The aim of the present study was to evaluate the use of fasting blood glucose (FBG) to predict ASCVD.

Methods: In all, 18 610 participants, aged 35–74 years at enrollment, were included in this prospective study. Baseline information was collected using a standardized questionnaire, physical examinations, and laboratory tests. During follow-up, disease status and vital information were updated. Cox proportional hazards regression analysis was used to estimate associations, with normal FBG (70–99 mg/dL) as the reference group. Anthropometric measurements, socioeconomic status, and conventional cardiovascular risk factors were included in the multivariate-adjusted model.

Results: After 7.8 years follow-up (145 223 person-years), there were 519 cases of ASCVD. The multivariate-adjusted hazard ratios (HR), with 95% confidence intervals (CI), for ASCVD in patients with low FBG (<70 mg/dL), impaired fasting glucose (IFG; 100–125 mg/dL), and diabetes (≥126 mg/dL, use of antidiabetic medication and/or self-report) were 1.35 (0.84, 2.15), 1.02 (0.81, 1.27), and 1.68 (1.26, 2.23), respectively. Although IFG was associated with the development of diabetes (multivariate-adjusted HR 3.67; 95% CI 3.20, 4.21), it was only associated with incident ASCVD in the univariate model (HR 1.52; 95% CI 1.23, 1.88). The association of diabetes with coronary heart disease was more pronounced than that with stroke. Gender and residential differences were also identified.

Conclusions: In the present study, IFG was associated with the development of diabetes but not incident ASCVD. Prevention strategies to reduce the development of diabetes in people with IFG are critical to improve cardiovascular health.

Keywords: cardiovascular diseases, diabetes mellitus, epidemiology, glucose.

Significant findings of the study: Diabetes rather than impaired fasting glucose was associated with an increased risk of incident ASCVD after 7.8 years follow-up among 18 610 participants.
What this study adds: Impaired fasting glucose increases the risk of developing diabetes and prevention strategies to reduce the development of diabetes in people with impaired fasting glucose are critical to improve cardiovascular health.

Introduction

Cardiovascular diseases (CVDs) have been the leading cause of death and years of life lost worldwide.[1] In recent decades, age-adjusted death rates attributable to CVDs have declined in developed countries, but have risen considerably in low- and middle-income countries, with cerebrovascular disease and ischemic heart disease as the top two leading causes of death in East Asia.[2,3] Therefore, identification of high-risk populations for CVDs is very important for target prevention.

Diabetes increases the risk of CVD,[4–6] but some uncertainties remain regarding the relationship between fasting blood glucose (FBG) levels in the normal range and CVD risk.[7] The American Diabetes Association (ADA) has lowered the limit of impaired fasting glucose (IFG) from 110 to 100 mg/dL based on the clinical benefits of diabetes prevention, but the benefits of primary prevention of CVD in populations with IFG remain contentious.[8–10] In addition, current knowledge about the association of FBG levels with CVDs is derived primarily from studies conducted in developed or high-income countries.[11,12] For example, the Korean Cancer Prevention Study has reported a J-shaped association between FBG levels and CVD risk,[12] as well as a U-shaped association with incident atherosclerotic cardiovascular disease (ASCVD).[13] However, evidence from developing countries, such as China, is limited.

A recent survey reported a great disease burden of diabetes and prediabetes in China, with a prevalence of diabetes and prediabetes in 2010 of 11.6% and 50.1%, respectively.[14] Results from a large cross-sectional study including 0.5 million people aged 30–79 years showed that self-reported diabetes and higher random blood glucose were associated with an increased risk of CVD[15] and that FBG was associated with angiographic coronary heart disease (CHD) in a suspected myocardial ischemia population.[16] One prospective study conducted in a Hong Kong Chinese population has identified HbA1c as a significant risk factor for stroke.[17] However, the generalizability of the results was limited by inclusion of only individuals with type 2 diabetes mellitus (T2DM).[17] Another prospective study with only 3 years follow-up from the Chinese mainland in 19 359 subjects aged 40–85 years identified that insulin resistance or a previous diagnosis of diabetes predicted first-ever stroke.[18] Therefore, further evidence from prospective studies is needed to demonstrate the use of FBG to predict ASCVD in Chinese populations. Taking into consideration limited healthcare resources, the identification of associations between FBG and ASCVD is critical to determine whether to conduct prevention strategies for ASCVD in populations with diabetes, IFG, or low fasting glucose (LFG) levels.

The present study was undertaken to examine the association of FBG levels with incident ASCVD over a 7.8-year follow-up period among Chinese individuals aged between 35–74 years and to provide evidence for the primary prevention of ASCVD in populations with different FBG levels.

Methods

Study population

Participants in the present study were drawn from the China Multicenter Collaborative Study of Cardiovascular Epidemiology (ChinaMUCA) study and the China Cardiovascular Health Study. Details of study design and inclusion criteria for both studies have been published elsewhere.[19,20] Briefly, the ChinaMUCA study included 15 clusters selected on the basis of socioeconomic status, geographical location, and dietary patterns, with approximately 1000 subjects aged 35–59 years in each cluster and approximately 50% of all participants being female. In the China Cardiovascular Health Study, a four-stage stratified sampling method was used to select a nationally representative sample of the general population aged 35–74 years. Baseline examinations were conducted in 1998 for the ChinaMUCA study and in 2000–01 for the China Cardiovascular Health Study.

In all, 27 020 individuals aged 35–74 years were included in the two studies (n = 11 480 from ChinaMUCA; n = 15,540 from the China Cardiovascular Health Study). All individuals were invited to participate in the follow-up study in 2007–08. After excluding 1854 individuals with missing baseline glucose information, 1039 individuals from Deyang, Sichuan, for whom information could not be confirmed because of an earthquake, and 501 individuals who had ASCVD at baseline, 23 626 participants were included in the follow-up study. The follow-up rate was 78.77%, with 5016 participants' lost to follow-up. Thus, 18 610 participants were included in the present analysis (Fig. 1). Written informed consent

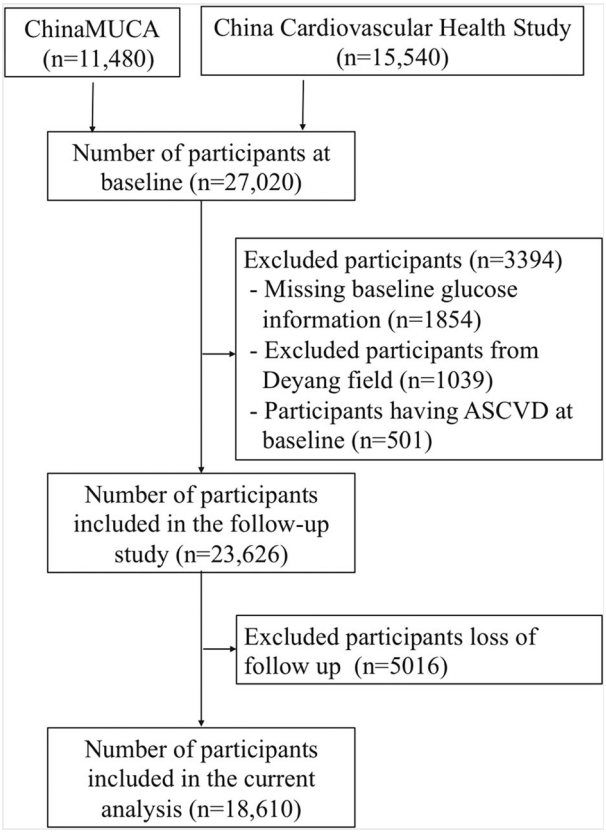

Figure 1 Selection of participants for examination of the association of fasting glucose levels with atherosclerotic cardiovascular disease (ASCVD). ChinaMUCA, China Multicenter Collaborative Study of Cardiovascular Epidemiology

was obtained from all participants after detailed explanation of the study. The study was approved by the Institutional Review Board at Fuwai Hospital.

Data collection

At the baseline visit, participants were interviewed and underwent physical examinations and laboratory tests. Information on demographic characteristics, personal and family medical history, and lifestyle risk factors was collected by trained staff using a standardized questionnaire. Body weight and height were measured to the nearest 0.5 kg and 0.5 cm, respectively, with participants in light indoor clothing without shoes. Body mass index (BMI) was calculated as kilograms per meter squared (kg/m^2). Blood pressure (BP) was measured three times in seated participants after 5 min rest by the same trained and certified technician according to a standardized procedure recommended by the American Heart Association.[21] Blood pressure levels were calculated as the average of the three BP measurements. Fasting blood samples were drawn from participants after at least a 10-h fast, and were centrifuged immediately at 2000*g* for 10 min at 4°C or room temperature. Blood samples were transferred under cold chain conditions to the Central Laboratory of the Department of Epidemiology, Fuwai Hospital (Beijing, China). This laboratory participates in the Lipid Standardization Program of the US Centers for Disease Control and Prevention. Serum glucose concentrations were measured by a modified hexokinase enzymatic method (automatic clinical analyzer, Model 7060; Hitachi, Tokyo, Japan) and total cholesterol (TC), high-density lipoprotein cholesterol (HDL-C), and triglycerides (TG) were assessed enzymatically using commercially available reagents.[22]

During the follow-up examination, study participants or their proxies were interviewed to ascertain disease status and other vital information. Hospital records and autopsy results, as well as death certificates from the local public health department or police department, were reviewed and verified by a study-wide endpoint assessment committee at Fuwai Hospital. Two committee members assessed the endpoints (occurrence of CVD event, cancer, and causes of mortality) independently and discrepancies were discussed with additional committee members. The committee members were unaware of subjects' baseline characteristics for risk factors. In the present analysis, ASCVD is defined as the first occurrence of non-fatal myocardial infarction (MI), CHD death, or fatal or non-fatal stroke.[23] Coronary heart disease events included a confirmed diagnosis of acute MI or death due to CHD, whereas stroke events included a confirmed diagnosis of stroke or deaths due to stroke during the follow-up period.

Statistical analysis

The characteristics of all study participants are presented as the mean ± SD or median values with the interquartile range (IQR) in parentheses for continuous variables and as percentages for categorical variables. Trend tests were conducted using a linear regression model with median value in each group for continuous variables and the Mantel-Haenszel Chi-squared test for categorical variables. To examine the effect of FBG levels on ASCVD risk, the study participants were divided into glucose groups: LFG (<70 mg/dL), normal fasting glucose (NFG; 70–99 mg/dL), IFG (100–125 mg/dL), and diabetes (≥126 mg/dL and/or using insulin and/or oral hypoglycemic agents, and/or a self-reported history of T2DM). The NFG group was used as the reference group. Hazard ratios (HRs) and their corresponding 95% confidence intervals (CIs) for ASCVD in each group were estimated by Cox proportional hazards

regression analysis in a univariate model, an age-adjusted model, and then in a multivariate-adjusted model with further adjustment for anthropometric measurements, socioeconomic status, and conventional cardiovascular risk factors. Furthermore, subgroup analyses were performed according to outcome, gender, region and area in China.

The definitions of related covariates included in the multivariate-adjusted model are as follows. Smoking was defined as having smoked at least 400 cigarettes or 500 g tobacco leaves throughout the lifetime, or having smoked at least one cigarette per day for 1 year or more. Drinking was defined as drinking alcohol at least once a week in the past 12 months. In addition, participants were divided into two groups according to work-related physical activity: none–mild or medium–high. Hypertension was defined as systolic BP (SBP) ≥140 mmHg and/or diastolic BP (DBP) ≥90 mmHg and/or currently using antihypertensive medicine. High TG was defined as TG ≥200 mg/dL.[24]

All analyses were performed using SAS 9.3 (SAS Institute, Cary, NC, USA). Two-sided $P < 0.05$ was considered significant.

Results

Table 1 presents the characteristics of the 18 610 participants according to glucose group. Overall, participants with higher FBG were older, more likely to have higher BMI, SBP, DBP, TC, or TG levels, and lower educational level, more likely to live in rural areas of China, and less likely to be current smokers. Over an average of 7.8 years follow-up (145 223 person-years), 519 incident ASCVD cases (134 CHD and 385 stroke cases) were identified, corresponding to an incidence rate of 3.57 per 1000 person-years for ASCVD. Compared with participants lost to follow-up, those followed-up were younger, more likely to be northern or rural residents, and had lower levels of BMI, better blood lipid status, and higher physical activity levels. In addition, the distribution of FBG status was a little different between the two groups ($P < 0.001$). For example, the proportion of participants with LFG was lower, whereas the proportion of those with IFG was higher in the lost to follow-up group (see Table S1, available as Supplementary Material to this paper).

The incidence rate of ASCVD (per 1000 person-years) in the LFG, NFG, IFG and diabetes groups was 2.93, 3.02, 4.50, and 8.80, respectively (Table 2). Figure 2 shows the Kaplan–Meier plot for the proportion of ASCVD event-free survival in each glucose group (Log rank $P < 0.001$). The HRs for ASCVD were significantly increased in participants with IFG (HR 1.52; 95% CI 1.23, 1.88; $P<0.001$) and diabetes (HR 2.97; 95% CI 2.27, 3.88; $P<0.001$) in univariate analysis, but only diabetic patients had a significantly higher risk of

Table 1 Baseline characteristics of the 18 610 participants according to fasting glucose levels

	LFG (n = 779)	NFG (n = 13220)	IFG (n = 3607)	Diabetes (n = 1004)	*P*-value
Glucose (mg/dL)	63.3±7.5	87.6±7.4	107.7±6.4	170.1±60.4	<0.001
Age (years)	45.9±7.3	47.6±8.9	50.9±9.7	52.8±9.5	<0.001
Males	437 (56.1%)	6222 (47.1%)	1787 (49.6%)	483 (48.1%)	0.919
BMI (kg/m^2)	22.2±3.0	23.3±3.4	24.2±3.6	25.1±3.9	<0.001
Smoking status					
Never smokers	445 (57.1%)	8272 (62.7%)	2276 (63.2%)	625 (62.3%)	<0.001
Former smokers	46 (5.9%)	751 (5.7%)	277 (7.7%)	95 (9.5%)	
Current smokers	288 (37.0%)	4166 (31.6%)	1051 (29.2%)	284 (28.3%)	
Alcohol-drinkers	220 (28.4%)	3204 (24.3%)	990 (27.5%)	236 (23.7%)	0.319
Medium–high work-related physical activity	408 (52.4%)	5373 (40.8%)	1337 (37.2%)	269 (26.9%)	<0.001
Urban area	168 (21.6%)	5318 (40.2%)	1576 (44.0%)	504 (50.2%)	<0.001
Northern China	309 (39.7%)	7147 (54.1%)	1908 (52.9%)	595 (59.3%)	<0.001
Education					
0–6 years	348 (44.8%)	5353 (41.0%)	1570 (44.3%)	438 (44.4%)	0.008
7–12 years	394 (50.7%)	6478 (49.7%)	1671 (47.1%)	459 (46.6%)	
≥13 years	35 (4.5%)	1216 (9.3%)	305 (8.6%)	89 (9.0%)	
SBP (mmHg)	117.7±17.9	122.2±19.0	128.3±20.9	132.7±21.6	<0.001
DBP (mmHg)	75.0±11.4	78.1±11.3	80.8±11.3	81.3±11.7	<0.001
TC (mg/dL)	183.4±34.4	183.7±36.5	195.6±39.4	204.5±48.3	<0.001
HDL-C (mg/dL)	53.6±13.2	51.7±13.1	50.4±13.9	47.4±23.7	<0.001
TG (mg/dL)	66.0 (89.0, 133.3)	77.0 (104.2, 145.1)	86.2 (122.2, 177.0)	100.0 (146.0, 219.4)	<0.001

Data are presented as the mean±SD or as the median (interquartile range) for continuous variables and as n (%) for categorical variables.
LFG, low fasting glucose; NFG, normal fasting glucose; IFG, impaired fasting glucose; BMI, body mass index; SBP, systolic blood pressure; DBP, diastolic blood pressure; TC, total cholesterol; HDL-C, high density lipoprotein cholesterol; TG, triglyceride.

Table 2 Baseline fasting glucose levels and risk of atherosclerotic cardiovascular disease, stroke and coronary heart disease in a Chinese population

	LFG (n = 779)	NFG (n = 13 220)	IFG (n = 3607)	Diabetes (n = 1004)
ASCVD				
No. events	20	315	120	64
Person-years	6830	104 466	26 654	7274
Incidence rate (per 1000 person-years)	2.93	3.02	4.50	8.80
Crude HR (95% CI)	0.96 (0.60, 1.49)	1.00	1.52 (1.23, 1.88)	2.97 (2.27, 3.88)
P-value	0.806	–	<0.001	<0.001
Age-adjusted HR (95% CI)	1.16 (0.74, 1.83)	1.00	1.14 (0.92, 1.42)	2.00 (1.52, 2.62)
P-value	0.523	–	0.216	<0.001
Multivariate-adjusted† HR (95% CI)	1.35 (0.84, 2.15)	1.00	1.02 (0.81, 1.27)	1.68 (1.26, 2.23)
P-value	0.212	–	0.897	<0.001
Stroke				
No. events	14	241	86	44
Person-years	6835	104 584	26 709	7310
Incidence rate (per 1000 person-years)	2.05	2.30	3.22	6.02
Crude HR (95% CI)	0.87 (0.51, 1.49)	1.00	1.42 (1.11, 1.82)	2.66 (1.93, 3.67)
P-value	0.614	–	0.005	<0.001
Age-adjusted HR (95% CI)	1.07 (0.62, 1.84)	1.00	1.07 (0.83, 1.37)	1.79 (1.29, 2.48)
P-value	0.800	–	0.606	<0.001
Multivariate-adjusted† HR (95% CI)	1.30 (0.75, 2.24)	1.00	0.96 (0.74, 1.25)	1.51 (1.07, 2.13)
P-value	0.346	–	0.960	0.019
Coronary heart disease				
No. events	6	74	34	20
Person-years	6870	104 974	26 818	7376
Incidence rate (per 1000 person-years)	0.87	0.70	1.27	2.71
Crude HR (95% CI)	1.19 (0.52, 2.75)	1.00	1.84 (1.23, 2.77)	3.90 (2.38, 6.40)
P-value	0.679	–	0.003	<0.001
Age-adjusted HR (95% CI)	1.46 (0.63, 3.36)	1.00	1.39 (0.92, 2.09)	2.62 (1.59, 4.33)
P-value	0.380	–	0.121	<0.001
Multivariate-adjusted† HR (95% CI)	1.49 (0.60, 3.72)	1.00	1.18 (0.77, 1.83)	2.17 (1.29, 3.63)
P-value	0.392	–	0.447	0.003

†Adjusted for age (every 10 years), gender, body mass index, smoking status (never smoker, former smoker, current smoker), drinking status (never drinker, drinker), work-related physical activity, education, hypertension status (yes vs no), high triglyceride (yes vs no), region (south vs north), area (urban vs rural), and family history of atherosclerotic cardiovascular disease (ASCVD).
HR, hazard ratio; LFG, low fasting glucose; NFG, normal fasting glucose; IFG, impaired fasting glucose.

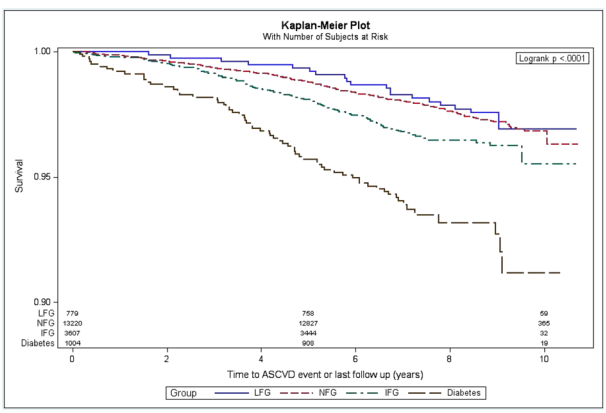

Figure 2 Kaplan-Meier curves of atherosclerotic cardiovascular disease (ASCVD) event-free survival of participants according to fasting (serum) glucose groups. LFG, low fasting glucose; NFG, normal fasting glucose; IFG, impaired fasting glucose. Values across the bottom of the graph show the number of subjects at risk at each time point.

developing ASCVD in age-adjusted and multivariate-adjusted models, with HRs of 2.00 (95% CI 1.52, 2.62; $P<0.001$) and 1.68 (95% CI 1.26, 2.23; $P<0.001$), respectively (Table 2; Fig. 3). When we analyzed the association of glucose with stroke and CHD separately, the crude HRs for stroke and CHD also increased significantly in the IFG and diabetes groups. However, after adjusting for age, gender, BMI, smoking status, drinking status, physical activity, education, family history of ASCVD, and other conventional cardiovascular risk factors, the increased HRs remained significant only in the diabetes group and the HR for CHD was more pronounced than that for stroke (Table 2). In addition, a potential J-shaped association was observed for the association between FBG levels and the incidence of ASCVD (Table 2).

During the follow-up period, 1034 participants without diabetes at the time of enrollment developed diabetes. Compared with NFG, IFG was associated with an

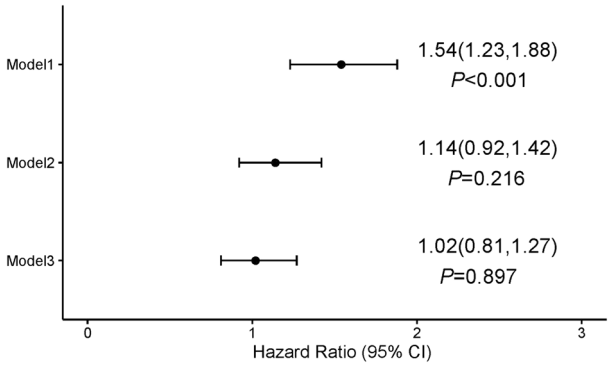

Figure 3 Risk of atherosclerotic cardiovascular disease (ASCVD) during the follow-up period in the group with impaired fasting glucose. Model 1, univariate model; Model 2, adjusted for age; Model 3, adjusted for age as well as gender, body mass index, smoking status, drinking status, work-related physical activity, education, hypertension, high triglyceride, region, area and family history of ASCVD. CI, confidence interval.

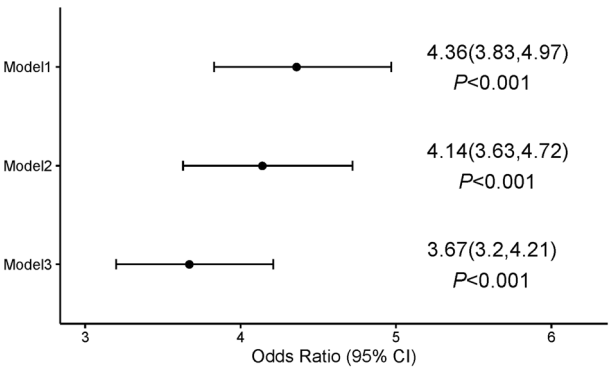

Figure 4 Risk of developing diabetes during the follow-up period in the group with impaired fasting glucose. Model 1, univariate model; Model 2, adjusted for age; Model 3, adjusted for age as well as smoking, drinking, physical activity, education, region (north vs south China) and area (urban vs rural).

increased risk of diabetes in the univariate model, age-adjusted model, and in the full model, with odds ratios (ORs) of 4.36 (95% CI 3.83, 4.97; $P<0.001$), 4.14 (95% CI 3.63, 4.72; $P<0.001$), and 3.67 (95% CI 3.20, 4.21; $P<0.001$), respectively (Fig. 4).

We conducted subgroup analysis with gender, region, or area with NFG as the reference group (Table 3). The diabetic population had a 40% and 105% increased risk of developing ASCVD for males and females, respectively. In addition, the HR (95% CI) for ASCVD among diabetic participants from northern China was significantly higher than for their counterparts from southern China, with multivariate-adjusted HRs (95% CI) of 1.66 (1.19, 2.33) and 1.61 (0.91, 2.51), respectively. However, the HR among diabetic participants in urban areas was similar to those for diabetic participants from rural area.

A sensitivity analysis in which participants developing ASCVD within 1 year of the baseline examination were excluded (n = 35) generated similar results to those given in Table 2 (see Table S2). Moreover, HRs for stroke and CHD did not change substantially when participants with history of diabetes were excluded (n = 357; Table S2).

Discussion

In this large cohort including both urban and rural Chinese individuals over 7.8 years of follow-up, diabetes was confirmed to be significantly associated with higher risk of developing ASCVD, stroke or CHD compared with FBG levels of 70–99 mg/dL. In addition, IFG was associated with an increased risk of developing diabetes rather than ASCVD. Aggressive lifestyle modifications and other preventive strategies for participants with IFG, which may reduce the risk of developing diabetes, also have an important effect in preventing ASCVD.

Previous studies found either an increased or decreased risk of CVD with diabetes or prediabetes in Western populations or populations from developed countries.[25–29] Several meta-analyses have found a positive association between IFG and CVD, but were limited by the lack of data for the Chinese population.[30–32] In addition, IFG is a confirmed risk factor for diabetes,[33] and the inclusion of studies with >20 years follow-up in the meta-analyses[30–32] may have biased the association. Individuals may have diabetes for at least 10 years in studies with such a long follow-up period, making it unclear whether the identified increased risk of CVD was due to diabetes or IFG.[31,34] Only two studies conducted in a Chinese population have investigated the ability of prediabetes to predict CVD, and hyperglycemia was found not to be associated with a significantly higher risk of CVD without any other concomitant disorders.[27,35] In addition, the association of LFG with CVD has not been well studied in the Chinese population. The present prospective study has confirmed the ability of diabetes to predict the development of ASCVD. In addition, the multivariate-adjusted model identified a potential association between LFG (<70 mg/dL) and ASCVD, but no association for the IFG group. However, only nine ASCVD events occurred in the LFG group; thus, further investigation with a longer follow-up duration is warranted into the association between LFG and ASCVD in the Chinese population.

Differences in the associations of FBG levels with ASCVD depending on gender, region (north vs south), and area (urban vs rural) in the present study suggest that evaluating the effect of hyperglycemia on the risk

Table 3 Multivariate-adjusted† hazard ratios (95% confidence intervals) for atherosclerotic cardiovascular disease according to gender, region (south vs north), and area (urban vs rural)

	LFG (n = 779)	NFG (n = 13 220)	IFG (n = 3607)	Diabetes (n = 1004)
Gender				
Male	1.34 (0.79, 2.27)	1.00	0.88 (0.65, 1.18)	1.40 (0.94, 2.07)
Female	1.11 (0.41, 3.02)	1.00	1.22 (0.86, 1.73)	2.05 (1.36, 3.11)
$P_{interaction}$	0.541	–	0.184	0.138
Region				
Southern China	1.55 (0.74, 3.26)	1.00	0.97 (0.65, 1.45)	1.61 (0.94, 2.76)
Northern China	1.31 (0.71, 2.41)	1.00	1.02 (0.78, 1.34)	1.66 (1.19, 2.33)
$P_{interaction}$	0.926	–	0.838	0.606
Area				
Rural area	1.69 (0.99, 2.89)	1.00	1.22 (0.92, 1.63)	1.55 (1.01, 2.37)
Urban area	1.01 (0.37, 2.76)	1.00	0.78 (0.53, 1.11)	1.78 (1.21, 2.63)
$P_{interaction}$	0.394	–	0.025	0.394

†Adjusted for age (every 10 years), gender, body mass index, smoking status (never smoker, former smoker, current smoker), drinking status (never drinker, drinker), work-related physical activity, education, hypertension status (yes vs. no), high triglyceride (yes vs. no), region (south, north), area (urban, rural), and family history of atherosclerotic cardiovascular disease (ASCVD); gender, region and area were not adjusted for in corresponding subgroup analyses.
LFG, low fasting glucose; NFG, normal fasting glucose; IFG, impaired fasting glucose.

of developing ASCVD in the total population is inappropriate. Given the higher risk in females and in participants from northern China or urban areas, more strict primary prevention strategies are needed for these groups. A positive age-adjusted association was observed for ASCVD, CHD, and stroke in male participants and participants from southern China or rural areas, but it was markedly attenuated in the fully adjusted analysis, whereas the associations in female participants, participants from northern China, and those from urban areas persisted. In addition, excluding those who developed ASCVD in the first year after baseline yielded similar results.

Although some studies support the association of IFG with the incidence of CVD events, the results remain contentious.[12,27,30] Wang et al. found IFG was not associated with incident CHD, with a relative risk of 0.89 (0.57–1.39).[35] Another study conducted in a Chinese population found that hyperglycemia was associated with increased risk of CVD only when other metabolic syndrome components were present.[27] These results imply that the association of IFG or diabetes with CVD may be driven in large part by other coexisting risk factors, including high BP and reduced HDL-C.[27] Yeboah et al. used the Multi-Ethinic Study of Atherosclerosis cohort to identify that IFG was a risk factor for diabetes but not CVD.[36] However, only 11.8% of the participants in that study were Chinese and race-specific analysis was not provided.[36] The present study provides similar evidence that diabetes is an independent risk factor for ASCVD and that IFG is associated with diabetes in a large population. The results revealed that IFG per se was not an independent risk factor for ASCVD, but the progression from IFG to diabetes may lead to the development of ASCVD.

At present, the mechanisms underlying greater ASCVD risk conferred by diabetes in females than males remain uncertain, but some potential reasons could include the fact that women with diabetes are more likely to have a worse cardiovascular risk profile,[37] and that, compared with their male counterparts, females are less likely to achieve target levels of cardiovascular risk factors.[38,39] In the present study, females with diabetes were older (52.8 vs 48.2 years) and had higher SBP (132.7 vs 123.3 mmHg) than men. In addition, the BMI of females with newly diagnosed diabetes during the follow-up period was 0.4 kg/m^2 higher than for their male counterparts.

The regional or area differences in the association of FBG levels with ASCVD could be explained, in part, by the following. In the present study, participants from northern China or urban areas had higher BMI, SBP, and DBP and lower HDL-C levels than those from southern China or rural areas, respectively. Differences in lifestyles, including a high-sodium diet or sedentary behavior, may have contributed to these differences.[40,41] However, because of the relatively shorter follow-up period and/or the relatively low incidence of ASCVD, the present study is underpowered to identify such associations in subgroup analyses.

We found the associations with FBG levels were stronger for CHD than for stroke, which is in line with results from a study conducted in Western populations.[42] However, the magnitude of the association was smaller in the Western population, because a higher reference level (FBG <140 mg/dL) was used than in the present

study (70–99 mg/dL), which meant that participants with high glucose levels were included in the reference group in the other study, which could have resulted in an underestimation of the strength of the association between FBG levels and stroke. Strokes account for a large proportion of ASCVD in populations from East Asia,[43] so considerable benefits will be seen with absolute reductions in stroke events, as well as CHD events, after specific prevention in individuals with diabetes. In addition, preventing progression of IFG to diabetes is a potential strategy to reduce the incidence of ASCVD.

However, some limitations of the present study need to be addressed. Because of limited ASCVD events in the LFG group, the relationship between LFG and ASCVD requires further investigation, with a longer follow-up period. In the present study, only FBG levels, rather than HbA1c or post-load glucose, was determined. However, determination of FBG levels is most cost-effective and can be widely used to screen populations with potential diabetes or at high-risk of CVD. In addition, the ADA recommends that FBG levels are determined in plasma,[44] whereas FBG levels were determined in the serum in the present study. Although serum glucose values are 1.15% lower than plasma glucose levels, this may not be physiologically relevant. In addition, all the samples were stored at –80°C after transportation under cold chain conditions, which minimized the decline in glucose levels.[45] Another limitation of the present study is that a single measurement of glucose levels may not be an accurate reflection of the effects of changes in blood glucose over time on the incidence of ASCVD. Furthermore, 20% of the study population was lost to follow-up, which may have affected the findings. However, participants included in the present analysis were much healthier, with a younger age, lower BMI levels, better blood lipid status, and higher physical activity levels than those lost to follow-up, and so the present study may have underestimated the effects of diabetes on the risk of ASCVD.

In conclusion, the findings of the present study suggest that individuals with diabetes rather than IFG are at increased risk of developing ASCVD. In addition, the ability to use diabetes to predict the development of ASCVD differed between regions and areas in the Chinese population evaluated herein, and the findings may encourage region- or area-specific approaches to ASCVD prevention. Together with the rapidly increasing number of people in China with diabetes or IFG, the findings suggest that preventive action is urgently needed to reduce incident diabetes in people with IFG and to further reduce the effect of diabetes on cardiovascular health.

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Disclosure

The authors declare they have no conflicts of interest.

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